

Quantum mechanics

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I find the typical explanations for the foundational assumptions of quantum mechanics to be lacking. Not many books take time to fully develop postulates, or axioms, from observations; not many books interleave observation, axiom, and derivation in the way I think should be the standard for physics. It's all too common for the postulate to come first, with its justification being a retroactive compare-and-contrast with classical theory.

Feynman's lecture notes on quantum mechanics come pretty close to what I hope for. Though, they don't derive everything, as I would hope for. They especially don't explain why wavefunctions are complex-valued. These notes are intended to fill that gap. The idea is that one reads this instead of Chapters 1 - 3 of Feynman, and then continues on with Feynman after this.

Classical prerequisites

History

The history of the question “what is light”?

- Late 1600s - early 1800s - Newton argues that light consists of light particles, while Huygens argues that light is a pulse¹ Until Young's double-slit experiment, Newton's theory won out due to his stature, and for how it better explained the fact that shadows had sharp edges. Though, Huygens notably proposed *Huygens's principle*, which states that every point on a wavefront can be treated as a new source emitting its own spherical wavelets, and that the wavefront at a later time is the envelope of all these wavelets.
- Early 1800s - Young performs his double-slit experiment, demonstrating that light exhibits wavelike interference. This upends Newton's theory.
- Early 1800s - Drawing on Huygens, Fresnel comes up with the theory for predicting the wave behavior of light in general settings that are more complicated than just a double-slit.
- Mid 1800s - Foucault uses a rotating mirror to measure the speed of light.
- Mid 1800s - Maxwell organizes together a set of equations for electromagnetism, adding a correction term of his own. He realizes (qualitatively) that accelerating charges must radiate sinusoidal electromagnetic waves, and that all electromagnetic waves travel at a speed very close to the speed for light that Foucault measured. Maxwell proposes that light must be an electromagnetic wave.
- *Monochromatic* light can now be defined to be a light wave that has a single wavelength. Only certain colors of light- the “ROY-G-BIV” colors of the rainbow- are monochromatic.
- Late 1800s - Hertz experimentally verifies Maxwell's radiation prediction.
- Late 1800s - Larmor derives the precise formula for the radiation of an accelerating charge.

The history of the spectrum, the atomic fingerprint

- Late 1600s - Newton splits white light into a continuous spectrum with a prism.
- Early 1800s - In the paper presenting his double-slit experiment, Young provides a means to calculate the wavelength of the fringes produced.
- Early 1800s - Fraunhofer observes the solar spectrum with a prism, and sees many black lines. He later uses a mechanical ruling engine to build the first high-quality diffraction gratings, and uses calculations inspired by Young to repeat his experiment more quantitatively.

¹Though Huygens used the word “wave”, his waves were not understood as decomposing into waves of single frequencies. Huygens in fact had no concept of wave interference.

- Mid 1800s - Kirchhoff and Bunsen establish that when heated in a flame and viewed through a prism, each element emits a characteristic spectrum. They start cataloging spectra for each element, and find that the black lines in the solar spectrum correspond to emission lines of certain elements.
- Late 1800s - Diffraction gratings are popularized and spectra are further cataloged.
- Late 1800s - The Balmer formula for the hydrogen spectrum and the Rydberg formula for alkali metal spectra are discovered.

Questions

- How do we know that matter is made of atoms? Wait, what even is an atom? Oh yeah, it's an indivisible piece of an element.
- What even is an element? Well, a *compound* is a substance with distinctive chemical properties consistent across all of its instances. (How does one even define "chemical property" before knowing about electrons? Is "chemical property" just vaguely "alchemical property"?) An *element* was one of the substances that makes up a compound with its own chemical properties and that cannot be separated, by the typical means of early chemistry, into other substances. (What were the "typical means of early chemistry"? Heating?) And an atom is a piece of an element that cannot be divided into smaller parts without sacrificing some of the element's chemical properties.
- J.J. Thomson noticed that the ratios of masses of elements found in compounds are (1) the same across all instances of a compound and (2) involve small integers. He thought (2) was most easily explained by there existing indivisible pieces of elements. Although I don't really buy this- it could have easily been the case that elements are infinitely divisible, and that some other principle or influence forced the ratios of small integers.
- Assuming that we do know that atoms exist, how do we know that their constituents have electrical charge, and are influenced by that force that affects amber rubbed with fur, electricity?
- What was the evidence for the plum pudding model of the atom, which depicted the atom as a positively charged sphere with small negatively charged particles (electrons) scattered uniformly throughout? In particular, how did we know that the negative charge wasn't continuously distributed, but instead in discrete particles?
- The periodicity of elements was known before electrons were, right?

Waves

Definition 1. (Frozen spatial wave).

A *transverse frozen spatial wave* is a function that describes the perpendicular displacement of every position on a straight line. Think of a snapshot in time of a wave traveling at sea. A *longitudinal frozen spatial wave* is a function that describes the parallel displacement of every position on a straight line. Think of a snapshot in time of a pulse traveling through a slinky.

Whether a frozen spatial wave is transverse or longitudinal, it has the form $(x, t) \mapsto A \cos\left(\frac{2\pi}{\lambda}x + \phi\right)$ for some $A, \lambda \geq 0$ and $\phi \in \mathbb{R}$.

Probably the most familiar type of wave is the transverse traveling wave. Although, longitudinal traveling waves also exist.

Definition 2. (Traveling wave). A *traveling wave* is a frozen spatial wave that is translated over time, and is thus a function of the form $(x, t) \mapsto A \cos\left(\frac{2\pi}{\lambda}x - \frac{2\pi}{T}t + \phi\right)$, where

- $\lambda \geq 0$ is the spatial period (the *wavelength*)
- $T \geq 0$ is the temporal period (the *period*), and $\nu := \frac{1}{T}$ is the temporal frequency (the *frequency*)
- $\phi \in \mathbb{R}$ is the phase shift (the *phase*)

Often, we use $k := \frac{2\pi}{\lambda}$, unilluminatingly called the *wavenumber*, and $\omega := \frac{2\pi}{T} = 2\pi\nu$ for brevity.

Definition 3. (Speed of a traveling wave). The *speed* v of a traveling wave is defined to be the time-rate at which it advances through a wavelength, $v := \frac{\lambda}{T} = \lambda\nu$.

In contrast to traveling waves, there is also the slightly less familiar standing wave. The best example of these is probably a vibrating string.

Definition 4. (Standing wave). A *standing wave* is a frozen spatial wave whose amplitude oscillates over time, and is thus a function of the form $(x, t) \mapsto A \sin(\omega t) \cos(kx + \phi)$.

Theorem 1. Every standing wave can be expressed as a sum of traveling waves.

But not all sums of traveling waves are standing waves. As you would expect, though, we still consider such sums to be waves. This in fact leads us to the general definition of a wave:

Definition 5. (Wave). A *wave* is sum of standing waves and traveling waves.

Consider a wave that is the sum of n traveling waves emitted from n different sources. We can write that wave as

$$(x, t) \mapsto \sum_{i=1}^n A_i \cos(k_i x - 2\pi\nu_i t + \phi_i).$$

Notice that there is in general no single temporal frequency associated with this wave, but instead the multiple temporal frequencies $\nu_1, \dots, \nu_n$.

Definition 6. (Frequency content of a wave).

We say that a wave *has the frequency* ν_i if ν_i appears as a temporal frequency, like in the above, in one of the sum terms for the wave.

Young's double-slit experiment

If we perform a double-slit experiment with water waves, a single wave emitted by a source will be split into two copies of itself by the barrier between the slits, so that each slit becomes an emitter. When Young repeated this experiment with light of a pure color, he saw identical interference patterns, which suggests light is a wave.

Theorem 2. (Grating equation).

In a double-slit experiment in which monochromatic light of wavelength λ is shone through slits separated by a distance d , maximum constructive interference occurs at the fringes whose diffraction angle θ satisfies $d \sin(\theta) = m\lambda$, where $m \in \mathbb{Z}$.

Theorem 3. (Wavelength from displacement).

The displacement of a fringe in the interference pattern produced from a diffraction grating determines its wavelength.

Proof. Let y and θ be the displacement and diffraction angle of a given fringe, and let L be the distance from the grating to the screen. Then $\tan(\theta) = y/L$. Solving for θ and substituting into the grating equation, we obtain $d \sin(\arctan(y/L)) = m\lambda$, which is equivalent to $d \frac{y}{\sqrt{y^2 + L^2}} = m\lambda$. $\square$

Fourier series and wave packets

See “Fourier series and wave packets.tex” and “Fourier series and wave packets.pdf”.

Dipole moments

[TO-DO]

The full double-slit experiment

There are some additional strange observations we can make in the double-slit experiment:

1. By turning the intensity of the source down, we observe that light is detected on the screen as discrete particles- nothing like a wave hitting a breakwater.
2. And if we have a single-photon source², we can further observe that exactly one photon is detected at a time.
3. Even with a low intensity source, the classical interference pattern still builds up over time.

We propose the following axioms to explain the observations:

Axiom 1. (Wave interference).

Light functionally travels like a wave that comes out of the emitter as one wave and is split by the “breakwater” into two identical waves that interfere with each other and sum to form an aggregate wave.

Axiom 2. (Discreteness).

The aggregate wave arrives as exactly one of the particles that would have constituted it.

Axiom 3. (Axiom: structured randomness).

The location of that particle is randomly chosen, such that the distribution of the probabilities of the possible locations is proportional to the intensity of the aggregate wave.

Definition 7. (Wavefunction).

The aggregate wave is usually called the *wavefunction*.

Since the intensity I of a classical electromagnetic wave is $I = A^2$, where A is the amplitude, this is equivalent to saying that the distribution of the possible locations is proportional to square of the amplitude of the wavefunction.

One more observation yields some more unexpected behavior. If we place photon detectors at each slit, the interference pattern disappears, and we only see brightness. To account for this, we propose another axiom:

Axiom 4. (Indistinguishable alternatives).

Only reemitted waves corresponding to indistinguishable alternative paths are included in the wavefunction.

²If we don’t have a single photon source, we can still deduce that one photon is detected at a time by placing a photon detector at each slit, and seeing that the two slits never both record a photon.

Let me briefly explain how the observation follows from this axiom. [TO-DO]. While there are potentially other axioms that could explain this most recent observation, this one seems relatively natural.

From light to mass

When a double-slit experiment is performed on electrons instead of light, we observe the same things as we did for light. This gives us what de Broglie concluded in 1924:

Axiom 5. (The strangeness affects particles with mass too).

The axioms from the previous page apply to not only to light, but also to particles with mass.

Atomic energies

The quantitative description of spectra empirically discovered by Rydberg in the late 1800s showed that many spectral frequencies are differences of numbers.

Axiom 6. (Rydberg formula).

Consider the spectrum of an element a from the first column of the periodic table. Each spectral line is of the form^a

$$\nu = S_a \left(\frac{1}{m^2} - \frac{1}{n^2} \right), \text{ where } S_a \geq 0 \text{ and } n, m \text{ are positive integers.}$$

^a $S_a = cRZ_a^2$, where c is the speed of light, R is the Rydberg constant, and Z_a is the atomic number.

In 1908, Ritz took this further and empirically found that for *every* element, there is a small set of numbers whose pairwise differences yield *all* of that element's spectral frequencies.

Axiom 7. (Ritz terms).

For the spectrum of every element a , there exists a relatively small set of numbers T_a (historically called *terms*) such that:

1. The frequency of every spectral line is of the form $T_2 - T_1$, where $T_1, T_2 \in T_a$.
2. Spectral lines of the form $T_1 + T_2$, $T_1, T_2 \in T_a$, are coincidental.

At this point in time, we knew that electrons are the part of the atom that emit light during spectrum collection³. So, we knew that observed spectral frequencies were somehow associated with the emission of light from electrons.

A natural idea is to interpret Rydberg's terms from T_a , which have units as frequency, as *literally* being frequencies. Thus, they too must somehow associated with the emission of light from electrons.

In 1905, Einstein made this association precise.

Axiom 8. (Energy depends on frequency).

There exists a constant $h \geq 0$ such that when a light particle or a particle with mass^a, described by a monochromatic wavefunction of frequency ν , hits the screen of the double-slit experiment with energy $E = h\nu$.

^aThe experiment was only done for light. The only good reason we have to think it holds for mass is because of the fact that mass experiences the strangeness that light does in the double-slit experiment.

³A spectrum was obtained by heating the desired element and then viewing with a prism or diffraction grating. The heating causes the atoms of the element to radiate, i.e. emit light. Since electron are have negligible mass relative to the nucleus, they are the parts of the atom that meaningfully accelerate during this process, and thus the parts of the atom that radiate during this process.

The situation thus became as follows:

- We observe energies associated with the radiation of light by electrons.
- These observed energies are differences of unobserved energies that are also associated with electrons.

Note: the electron is the focus here, not the light emitted by it. I'm not sure why we can just ignore the quantum behavior of light here... but that is what Bohr did. And actually, Bohr was even able to make this leap without knowing that matter experiences the effects of the double-slit experiment!

Bohr hypothesized that each unobserved energy is a possible energy value that an electron can attain, and that, by conservation of energy, each observed energy must be a change in some electron's energy.

Axiom 9. (Electron energies).

The set of energy values that an electron bound to an atom a can attain is $\{ht \mid t \in T_a\}$, where T_a is the set of Ritz terms for the spectrum of a .

Notably, this meant that the energy of an electron was a member of a finite set. Energy became *quantized*.

Bohr's assumption makes a lot of sense at first glance, but when think more about Ritz's term structure, it very quickly becomes weird. Since observed frequencies are differences of unobserved frequencies, we can shift the unobserved frequencies around without changing the observed ones. This doesn't really make sense! One would think that an unobserved frequency of 0, which intuitively corresponds to zero oscillation, should in no way be allowed to be relabeled as a frequency of S , which corresponds to nonzero oscillation. But apparently not. Apparently, whether an unobserved thing is oscillating or not is a matter of perspective.

Indeed, we discover that the unobserved frequencies are in fact the frequencies of the electron's wavefunction, which cannot be observed...

Theorem 4. Since we observe frequencies that are differences of unobserved frequencies, the wavefunction of the electron must contain each of those frequencies individually.

Proof. The frequencies we observe are the frequencies appearing in dipole moment, $d(t) = e \int_{-\infty}^{\infty} x \rho(x, t) dx$. By the Born rule we have $\rho(x, t) = A(x, t)^2$. And since A is a wavefunction, a sum of traveling waves, we have

$$A(x, t) = \sum_{k=1}^n A_k(x) \cos(2\pi\nu_k t + \phi_k(x)),$$

for some positive integer n . Therefore $\rho = A^2$ is of the form

$$\rho(x, t) = \rho_0(x) + \sum_{k=1}^m \rho_k(x) \cos(2\pi\nu_k t + \tilde{\phi}_k(x)),$$

for some positive integer m . Now we pass from observation, to $d(t)$, to $\rho(x, t)$, to $A(x, t)$. We observe the frequency $T_2 - T_1$. Therefore $d(t)$ has $T_2 - T_1$ as a frequency. Now, since integrating a sinusoid against space cannot introduce new temporal frequencies, we can see that if a frequency does not appear in $\rho(x, t)$, it will not appear in $d(t)$. Thus $\rho(x, t)$ has $T_2 - T_1$ as a frequency.

Looking at the above expression for $A(x, t)$, we can see that if both of T_1 and T_2 do not appear individually as frequencies in $A(x, t)$, then the cross term necessary to produce a frequency of $T_2 - T_1$ will not appear as a frequency in $\rho(x, t) = A(x, t)^2$. So $A(x, t)$ has T_1 and T_2 as frequencies. $\square$

Theorem 5. If the wavefunction of an electron in an atom of element a has T_1 and T_2 as frequencies, then we should observe both $T_2 - T_1$ and $T_2 + T_1$ as frequencies in the spectrum of a .

Proof. Assume the electron wavefunction has both of T_1 and T_2 as frequencies. Then, using the expression for $A(x, t)$ from the previous proof, we see that the electron wavefunction must be given by

$$A(x, t) = A_1(x) \cos(2\pi T_1 t + \phi_1(x)) + A_2(x) \cos(2\pi T_2 t + \phi_2(x)) + \sum_{k=1}^r A_k(x) \cos(2\pi \nu_k t + \phi_k(x)),$$

for some positive integer r .

Now, in general, if $(x, t) \mapsto A_i(x, t)$ is a set of traveling waves, then any frequencies present in a squared sum of some of the waves $\left(\sum_{i=1}^p A_i(x, t)\right)^2$ will also be present in a squared sum of more of the waves $\left(\sum_{i=1}^q A_i(x, t)\right)^2$, $q > p$. It therefore suffices to assume that the third term in the above is zero, and that

$$A(x, t) = \underbrace{A_1(x) \cos(2\pi T_1 t + \phi_1(x))}_{A_1(x, t)} + \underbrace{A_2(x) \cos(2\pi T_2 t + \phi_2(x))}_{A_2(x, t)}.$$

Now we pass from this $A(x, t)$, to $\rho(x, t)$, to $d(t)$, to observation. First, we compute $\rho(x, t)$ by using Born's rule and the identities $\cos^2(\theta) = \frac{1}{2}(1 + \cos(2\theta))$, $2\cos(\alpha)\cos(\beta) = \cos(\alpha - \beta) + \cos(\alpha + \beta)$. We have

$$\begin{aligned} \rho(x, t) &= A(x, t)^2 \\ &= \underbrace{\frac{1}{2}(A_1(x)^2 + A_2(x)^2) + \frac{1}{2}A_1(x)^2 \cos(2\pi \cdot 2T_1 t + 2\delta_1) + \frac{1}{2}A_2(x)^2 \cos(2\pi \cdot 2T_2 t + 2\delta_2)}_{A_1(x, t)^2 + A_2(x, t)^2} \\ &\quad + \underbrace{A_1(x)A_2(x) \cos(2\pi(T_1 - T_2)t + \delta_1 - \delta_2) + A_1(x)A_2(x) \cos(2\pi(T_1 + T_2)t + \delta_1 + \delta_2)}_{2A_1(x, t)A_2(x, t)}. \end{aligned}$$

(Notice, five frequencies are present. The first line of the above has 0, $2T_1$, and $2T_2$, and the second line of the above has $T_1 - T_2$ and $T_1 + T_2$.)

Now, from this probability distribution, we compute the dipole moment:

$$\begin{aligned} d(t) &= e \int_{-\infty}^{\infty} x \rho(x) dx \\ &= e \int_{-\infty}^{\infty} x (A_1(x, t)^2 + A_2(x, t)^2) dx + e \int_{-\infty}^{\infty} x (2A_1(x, t)A_2(x, t)) dx. \end{aligned}$$

The first integrand is an odd function; thus, the first integral is zero by symmetry, and the frequencies associated with its integrand (0, $2T_1$, $2T_2$) do not appear. The second line integrand has the frequencies $T_1 - T_2$ and $T_1 + T_2$. Since the integration of a non-odd sinusoid against space preserves temporal frequencies^a, the dipole moment should have the frequencies $T_1 - T_2$ and $T_1 + T_2$.

Thus, we should observe spectral lines of frequencies $T_1 - T_2$ and $T_1 + T_2$. □

^aThe integration of an odd sinusoid against space is zero.

Except, we know from the Ritz axiom that we only observe the spectral lines whose frequencies are differences.

This only shows the problem with real-valued wavefunctions... and doesn't explain why complex-valued wavefunctions work. But anyhow, at this point, if you use the simplest and most obvious complex-valued wavefunction that contains T_1 and T_2 as frequencies, it works!

TO-DO

- $\mathbf{p} = \hbar \mathbf{k}$
- (Heisenberg uncertainty principle). We know from Fourier analysis that $\Delta x \Delta k \geq \frac{1}{2}$. It follows from the previous item that $\Delta x \Delta p \geq \frac{1}{2}$.